

Topic 2.6 - Competing Function Model Validation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Model selection is an argument supported by structure, not a contest to find the largest correlation number. Strong validation combines a plausible mechanism, a well-scaled residual analysis, comparison with competing models, and an explicit statement of the intended domain.

Learning targets

- Select plausible linear, quadratic, or exponential models from patterns, context, and residual behavior.
- Compute and interpret signed residuals and absolute error measures.
- Validate a model within a domain and articulate consequences of overprediction, underprediction, and extrapolation.

Language and structure

Term	Meaning in this topic
residual	Observed value minus predicted value: $r = y - \hat{y}$.
underprediction	A positive residual; the model prediction is below the observed value.
overprediction	A negative residual; the model prediction is above the observed value.
mean absolute error	The average of the absolute residuals.
systematic pattern	A nonrandom residual trend suggesting that important structure remains unmodeled.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Three models fit the same five data points: a linear model with MAE 2.8, a quadratic model with MAE 1.1, and an exponential model with MAE 1.3. What additional evidence is needed before selecting the quadratic model?

Concept 1: Choose candidates from structural fingerprints

Core idea

Linear data have constant first differences, quadratic data have constant second differences, and exponential data have constant ratios over equal input intervals.

- Approximate patterns suggest candidates but do not prove a unique model.
- Context can eliminate a mathematically good fit that contradicts the mechanism or allowable outputs.
- Use the same input interval when comparing differences or ratios.

Worked example - annotate the reasoning

Problem. The outputs 10, 15, 22, 31, 42 correspond to inputs 0, 1, 2, 3, 4. Identify a plausible exact model.

Model reasoning. First differences are 5, 7, 9, 11 and second differences are constantly 2, so a quadratic model is appropriate. Solving gives $y = x^2 + 4x + 10$.

Check your understanding

Independent

The outputs are 96, 72, 54, 40.5, 30.375 at consecutive inputs. Identify a plausible model.

Concept 2: Compute and read residuals

Core idea

A residual is $r = y - \hat{y}$; its sign describes the direction of model error.

- Positive residual: the model underpredicts.
- Negative residual: the model overpredicts.
- Small residuals scattered around 0 support a model more than residuals with a curved or one-sided pattern.

Worked example - annotate the reasoning

Problem. For observed values 80, 64, 51, 41, 33 at $x = 0, 1, 2, 3, 4$, compare $E(x) = 80(0.8)^x$ with $L(x) = 80 - 12x$ using residuals.

Model reasoning. For E , predictions are 80, 64, 51.2, 40.96, 32.768 and residuals are 0, 0, -0.2, 0.04, 0.232. For L , predictions are 80, 68, 56, 44, 32 and residuals are 0, -4, -5, -3, 1. The exponential model has much smaller and less patterned residuals.

Check your understanding

Independent

A model predicts 52 when the observed value is 47. Find and interpret the residual.

Concept 3: Validate within a domain

Core idea

A model can be useful on one interval and unacceptable outside it.

- Compare multiple error measures and inspect residual plots rather than relying on a single statistic.
- State consequences: underpredicting demand, cost, or risk may be more serious than overpredicting.
- Extrapolation requires the mechanism and constraints to remain stable.

Worked example - annotate the reasoning

Problem. A demand model has MAE 4.1 on weeks 1–12 and MAE 18.7 on weeks 13–16. What conclusion is justified?

Model reasoning. The model is reasonably accurate on the calibration interval but degrades substantially later. It should not be treated as validated for weeks 13–16 without revision or new evidence.

Check your understanding

Independent

Why might a slightly less accurate model be preferred to the smallest-MAE model?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

If $y = 31$ and $\hat{y} = 35.5$, compute and interpret the residual.

Exit ticket 2

No calculator

What residual pattern suggests that a linear model is missing curvature?

Exit ticket 3

No calculator

Why should a model's validation domain be stated?
